

PEPS Gauge Uniqueness and Balanced Edge Scalars

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1 The assertion

The uniqueness clause in the Fundamental Theorem for injective PEPS on a simple graph appears in [MGRPG⁺18, Theorem 2 and Section 3]. The argument first blocks the graph around a chosen edge to a three-site injective MPS. The one-dimensional fundamental theorem then assigns a gauge to that edge. Repeating this construction over all edges gives a family of edge gauges, and a local two-tensor comparison gives scalar proportionalities at the vertices.¹

The theorem is then stated for PEPS on graphs with no double edges and no self-loops. It says that two injective PEPS generate the same state if and only if their tensors are related by local gauges. In short, the source says that the local gauges are “unique up to a multiplicative constant.”² Interpreted as uniqueness up to one common scalar multiplying all edge gauges, this statement is too small for a general graph.

2 The corrected quotient

Let $G = (V, E)$ be a finite simple graph with a fixed orientation of each edge. For an edge e , a local gauge family assigns an invertible matrix X_e on the bond space of e . At the tail of e the endpoint action is X_e , and at the head of e it is X_e^{-1} .

A family of nonzero edge scalars $c = (c_e)_{e \in E}$ determines endpoint scalars

$$c_{v,e} = \begin{cases} c_e, & \text{if } v \text{ is the tail of } e, \\ c_e^{-1}, & \text{if } v \text{ is the head of } e. \end{cases}$$

¹In the repository source, the edge-blocking step is at `Papers/1804.04964/paper_normal.tex:981--1009`; the assignment of edge gauges is at `Papers/1804.04964/paper_normal.tex:1037--1065`; the local scalar proportionality argument is at `Papers/1804.04964/paper_normal.tex:1066--1092` and `Papers/1804.04964/paper_normal.tex:1204--1207`.

²The quoted uniqueness phrase occurs in `Papers/1804.04964/paper_normal.tex:1207--1210`. For traceability, this note corresponds to GitHub issue #1045.

The family is vertex-balanced if

$$\prod_{e \ni v} c_{v,e} = 1 \quad \text{for every } v \in V.$$

Two gauge families X and Y are equivalent modulo balanced edge scalars if there is such a vertex-balanced family c with

$$X_{v,e} = c_{v,e} Y_{v,e} \quad \text{for every incident pair } (v, e),$$

where $X_{v,e}$ and $Y_{v,e}$ denote the oriented endpoint actions of the two families.

This is the graph-correct quotient. Rescaling the incident endpoint gauges at a vertex v by the factors $c_{v,e}$ multiplies the gauged local tensor at v by $\prod_{e \ni v} c_{v,e}$. If the scalar family is vertex-balanced, this product is 1 at every vertex, so all local gauged tensors are unchanged. Thus local gauge uniqueness can at best be uniqueness modulo these balanced edge scalars. The formal statement uses this quotient as the corrected uniqueness statement; the quotient relation is an equivalence relation on gauge families.³

3 The triangle witness

The obstruction already appears on the connected triangle. Let the vertices be $1 < 2 < 3$, orient every edge from the smaller endpoint to the larger endpoint, and let every bond dimension be 1. Then an edge gauge is just a nonzero complex number. Take any nonzero injective bond-dimension-one PEPS tensor A , and compare the identity gauge family with the scalar gauge family

$$c_{12} = t, \quad c_{23} = t, \quad c_{13} = t^{-1}, \quad t \in \mathbb{C}^\times, \quad t^2 \neq 1.$$

At the three vertices the oriented products are

$$c_{12}c_{13} = 1, \quad c_{12}^{-1}c_{23} = 1, \quad c_{13}^{-1}c_{23}^{-1} = 1.$$

Hence the family is vertex-balanced. Applying this gauge family to A gives back the same local tensor at every vertex, so it is a second gauge family relating A to itself.

There is no single global scalar s with $c_{12} = c_{13} = c_{23} = s$. Indeed, $c_{12} = c_{13}$ would give $t = t^{-1}$, contradicting $t^2 \neq 1$. Therefore the global-scalar uniqueness interpretation is false even for a connected simple graph with one-dimensional bonds. The same example fits the balanced edge-scalar quotient exactly.⁴

³The formal definitions are `TNLean.PEPS.edgeScalarAt`, `TNLean.PEPS.isVertexBalanced`, and `TNLean.PEPS.GaugeEquivModEdgeScalars`; the quotient relation is `TNLean.PEPS.GaugeEquivModEdgeScalars.setoid` in `TNLean/PEPS/FundamentalTheorem.lean`. The corresponding blueprint entries are in `blueprint/src/chapter/ch13a_peps_ft.tex`.

⁴This is the counterexample recorded in `docs/counterexamples.md:127--135`. It originated in GitHub issue #762; the formal statement was corrected in PR #790.

4 Formal statement

The discrepancy is a statement-level divergence, not only an unfinished proof. The triangle example satisfies the graph-theoretic setting of the uniqueness clause and produces two genuine gauge families which differ by a balanced edge-scalar family but not by one global scalar. Consequently the source phrase must either be read with this quotient understood, or corrected to say uniqueness modulo balanced edge scalars.

The corrected quotient has not yet been fully proved. The repaired uniqueness result is stated formally, but its proof still has a remaining tensor-factor scalar-ratio and global-reconciliation step. The formal proof has reduced the problem to equality of products of incident edge-gauge entries at each vertex; what remains is to extract nonzero scalar ratios on the incident edges with product 1 at the vertex, and then reconcile these local choices into a single global vertex-balanced edge-scalar family.⁵

Thus the corrected mathematical statement is uniqueness modulo balanced edge scalars, and the triangle example refutes the smaller quotient. The repaired uniqueness theorem remains a stated formal result rather than a completed proof.

References

- [MGRPG⁺18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.

⁵For traceability, the declaration is `TNLean.PEPS.gauge.unique.mod.edge.scalars`, and the remaining proof obligation is at `TNLean/PEPS/FundamentalTheorem.lean:686--730`. This point is recorded in GitHub issue #842; PR #879 reduced the formal statement to the scalar-ratio and global-reconciliation step.